

# Madhan Senthilkumar

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## PROFESSIONAL SUMMARY

Full-stack software engineer and AI/ML developer with 2 internships and 3 deployed projects. Built and shipped a production MERN application with role-based access, real-time Firebase sync, REST APIs, and geospatial routing. Proficient in JavaScript, TypeScript, React.js, Node.js, Python, and SQL. Solved 100+ LeetCode problems across arrays, trees, graphs, and dynamic programming. Experienced in deploying scalable web applications and integrating machine learning models into production systems.

## EDUCATION

B.Tech in Artificial Intelligence and Machine Learning – Parul University, Vadodara CGPA: 8.30 | Aug 2023 – 2027

## TECHNICAL SKILLS

|                   |                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------|
| Programming       | Java, Python, JavaScript, TypeScript, SQL                                                     |
| Web Development   | React.js, Next.js, Node.js, Express.js, REST APIs                                             |
| Machine Learning  | Supervised Learning, Classification, Regression, Random Forest, CNN, TensorFlow, Scikit-learn |
| Data & Database   | Pandas, EDA, MongoDB, MySQL, Firebase, SQL Querying                                           |
| Core CS Concepts  | Data Structures & Algorithms, OOP, OS Fundamentals, DBMS                                      |
| Tools & Platforms | Git, GitHub, Docker, Postman, Vercel, Jupyter Notebook                                        |

## PROJECTS

SafeGo – Inclusive Ride Assistance & Safety Platform [GitHub] MERN, Firebase, OpenStreetMap, OSRM, ML | 2025

- Architected and deployed a full-stack MERN application with role-based access control, multilingual modules, and real-time admin monitoring – supporting PWD, women, elderly, and general users across distinct workflows.
- Designed and integrated RESTful APIs for live route navigation, distance calculation, and ETA tracking using OpenStreetMap and OSRM; implemented real-time data sync and authentication via Firebase.
- Embedded Random Forest Classifier and Regressor for geospatial safety scoring, integrating ML predictions directly into the production backend via a Python inference layer.

DataSanitizer Pro [Live] Next.js, React, TypeScript, Tailwind CSS | 2025

- Built and deployed a full-stack Next.js application that auto-detects missing values, duplicates, outliers, and inconsistencies in CSV/XLSX datasets with per-column quality reports and real-time previews.
- Implemented scalable data ingestion, server-side validation, and cleaned-data export pipelines; deployed on Vercel with a responsive TypeScript/Tailwind UI.

Flood Detection Using Deep Learning [GitHub] Python, TensorFlow, CNN, Image Processing | Sep 2024 – Oct 2024

- Built a CNN image classifier to detect flood-affected regions from satellite imagery using data augmentation and preprocessing, achieving ~85% classification accuracy.
- Automated before-and-after satellite image comparison to identify submerged land areas, supporting early disaster warning systems.

## EXPERIENCE

Data Science Intern Future Interns | Dec 2025 – Jan 2026

- Designed a weighted scoring model across 5-point Likert scales to quantify student satisfaction, identifying a top-performing metric at 4.21/5 and flagging 3 critical areas below 3.6/5 for institutional review.
- Delivered an interactive visual report using Pandas and Seaborn translating raw survey responses into actionable recommendations for 2 key improvement areas.

Machine Learning Intern Cognifyz Technologies (Remote) | Nov 2024 – Dec 2024

- Designed and executed end-to-end EDA and preprocessing pipelines using Python and Pandas on large-scale datasets, improving data reliability for downstream modeling.
- Built and evaluated multiple classification and regression models using Scikit-learn, selecting the best-performing model based on accuracy, precision, and F1-score metrics.

## CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Developer – Associate (Infosys Springboard)
- Oracle – Agentic AI Certified Foundations Associate
- Anthropic – Claude Code 101
- HackerRank – SQL (Advanced) Certificate
- IBM – Python for Data Science (2025)